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The Effects of Electronic Glossing Types on Foreign Language Vocabulary Learning: Different Types of Format and Glossary Information

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Abstract

This study investigated the effects of three different types of electronic textual glosses, tooltip-type glossing, frame-type glossing, and concordance glossing on foreign language vocabulary learning. While the first two glossing types provide the definitions of glossed words, with the only difference being their user interface designs, the third type provides authentic concordance sentences for glossed words. A total of 83 university students of English as a foreign language (EFL) participated in the study. They completed a computer-based reading task, meaning-recall vocabulary tests at three different points in time, and a post-reading questionnaire. The results showed that the intermediate EFL learners were affected not by a difference in terms of glossing formats, but by the type of information provided, with tooltip-type and frame-type glosses bringing about more positive learning outcomes. While the three groups all experienced a similar cognitive load, the findings further revealed that the tooltip-type and frame-type groups made greater gains of target vocabulary than their concordance counterpart. Moreover, the participants from the two former groups consequently rated their respective glossing more positively than those from the latter group.

Keywords: Computer-assisted language learning · Concordance · Electronic glossing · Frame-type glossing · Glossing format · Tooltip-type glossing · Vocabulary acquisition

Introduction

Vocabulary learning through reading has often been presented as a primary method of acquisition of both first language (L1) (Nagy et al. 1985) and foreign language (FL) lexical items (Krashen 1989). However, it may not be so effective for those who study FL in an instructional context with a limited amount of study time, as this process is generally found to be "slow" (Paribakht and Wesche 1997, p. 174). In response to this limitation, researchers have begun to explore possible ways to accelerate the process of vocabulary acquisition, while not sacrificing meaning-centered tasks (e.g., communicative activities, the reading of FL texts). Based on the results from previous empirical studies, glossing—supplementary lexical information provided in reading texts—has been suggested in FL reading research as an effective and convenient device to help FL learners read texts that are linguistically challenging at their current proficiency level as well as to enhance their learning of unfamiliar words (Schmitt 2008).

Since computer technology has become almost ubiquitous in classrooms, the pedagogical role of glosses in hypermedia environments has increasingly gained the attention of the research community (Abraham 2008). Glosses in hypermedia environments (electronic glosses, henceforth) distinguish themselves from printed glosses in terms of interaction between learners and computers mediated by technology-based learning environments via hypermedia properties. That is, "technology-mediated" environments "provide controlled opportunities for linguistic input for the learner and interaction with the computer" (Chapelle 2003, p. 25), while learners are exposed to printed glosses regardless of their intention.

Hypermedia environments can add yet another dimension to glosses: Different formats of electronic glosses can be constructed and realized in such environments. For example, electronic glosses may have several different formats in terms of their physical locations on users' screens or the conditions under which they can be activated or switched off. Furthermore, these formats may be expanded to include various types of glossary information. This would mean that glossary information such as concordance lines (Johns 1986) could easily be integrated into hypermedia environments, which otherwise would be difficult to do in print form due to the spatial limitations inherent in printed texts.

The present study was driven by Nation's (2009) introduction to the different types of glossing available for enhancing FL vocabulary learning in computer-assisted learning environments. According to his and others' descriptions (e.g., Al-Seghayer 2003; Lee and Lee 2013), different types of glossing may include different *formats*, which refer to the physical form and location of glosses projected on the screen, or different types of *glossary information* such as the definition of a glossed word and example sentences of a glossed word extracted from a

corpus (i.e., concordance lines). In particular, Nation (2009) and Al-Seghayer's (2003) descriptions of different types of glossing have recently engaged the interest of FL vocabulary researchers, with one strand of research examining different glossing formats and another investigating different types of glossary information. However, there have been few efforts to bridge these two strands together. To this end, the aim of the present study is to compare the effects of three different electronic glossing types—namely tooltip-type, frame-type, and concordance glosses—which have attracted increasing attention in recent studies (e.g., AbuSeileek 2011; Nation 2009; Poole 2012), but which have not been compared systematically in one empirical study in terms of their effects on FL vocabulary learning.

Literature Review

With the development of ever-more-rapidly changing technology, research into FL reading and vocabulary quickly entered the realm of computer-assisted language learning (CALL). In this research area, glossing has been demonstrated to enhance reading comprehension as well as FL vocabulary acquisition, as found in a meta-analysis of studies on electronic textual glosses (Abraham 2008). While previous studies have uncovered significant evidence in favor of the positive effects of electronic glossing, research inquiries into which glossing format and type of information would facilitate vocabulary acquisition have only recently gained attention from researchers, which will be discussed in what follows.

The Effects of Different Glossing Formats

One strand of inquiry that requires more attention in glossing research is where and how glossing should be placed for FL learners (Al-Seghayer 2003). In this line of research, Yeung (1999) examined the effects of two different glossing formats in a paper-based reading context —(1) an integrated format of glossing where vocabulary definition was given right above the glossed word and (2) a separated format which referred to a list of target words and their definitions given at the end of the text, in terms of the reading comprehension, vocabulary learning, and imposed cognitive load on learners with different FL proficiencies. The study found that the learners who were at a lower level of proficiency benefited more from the integrated format in terms of reading comprehension, but not in terms of vocabulary learning; the reverse pattern was found with the learners who were at a higher level of proficiency.

In a hypermedia environment, AbuSeileek (2011) compared the effects of different formats of electronic textual glosses on the vocabulary acquisition of undergraduate English as foreign language (EFL) learners. In this study, he designed different glossing locations in a hypermedia reading environment, such as (1) marginal glosses, which presented glossary information beside the glossed word in the margin, (2) pop-up window glosses, which

appeared in a box format when users clicked the glossed word, (3) bottom glosses, which appeared at the bottom of the screen when activated, and (4) after-glossed-word glosses, in which the glossary information was inserted into the text directly after the glossed word, as the reader clicked it. The results showed that the fourth group performed significantly better than the other groups, whereas the 'marginal' group outperformed the 'bottom' group on a multiple-choice vocabulary test. On the other hand, in the EFL context of Taiwan at the tertiary level, Chen and Yen (2013) made a comparison of the following formats of electronic textual glosses: (1) in-text annotation, where glossary information was typed right after the target word as part of the text; (2) pop-up glosses; and (3) glossary list at the end of the text. Interestingly, the three treatment conditions made comparable gains in terms of target vocabulary, suggesting that the results did not correspond to those of AbuSeileek (2011).

The results of a limited number of studies of glossing format are far from conclusive. In particular, inconsistent findings between Chen and Yen (2013) and AbuSeileek (2011) need further empirical investigation before we suggest one particular type of glossing format over another to FL learners. In light of previous studies, the present study aims to introduce and compare tooltip-type and frame-type glosses (see Lee and Lee 2013 for technical discussions). These two formats have been constructed in light of the suggestions that Al-Seghayer (2003) made in terms of how electronic glosses should be displayed (see Appendix 1 for details on them).

The rationale behind researching these two formats of electronic glossing arises from both theoretical needs and practical considerations. First, there has been no previous study of FL contexts that has examined the effects of tooltip-type glosses in comparison with other types of glosses (frame-type glosses in this case). Second, these two formats are arguably the most widely used forms of electronic textual glosses today; thus, it is of considerable importance to investigate their relative effectiveness for our FL learners. Amazon's Kindle, for example, provides users with a dictionary function which presents the definitions of words on the bottom of the screen. This dictionary function is similar to frame-type glosses in the present study. On the other hand, a dictionary function that resembles tooltip-type glosses is included in the mobile version of Safari on the iPad, another popular digital device today. In this device, Safari provides definitions of words right below them (see Fig. 1).

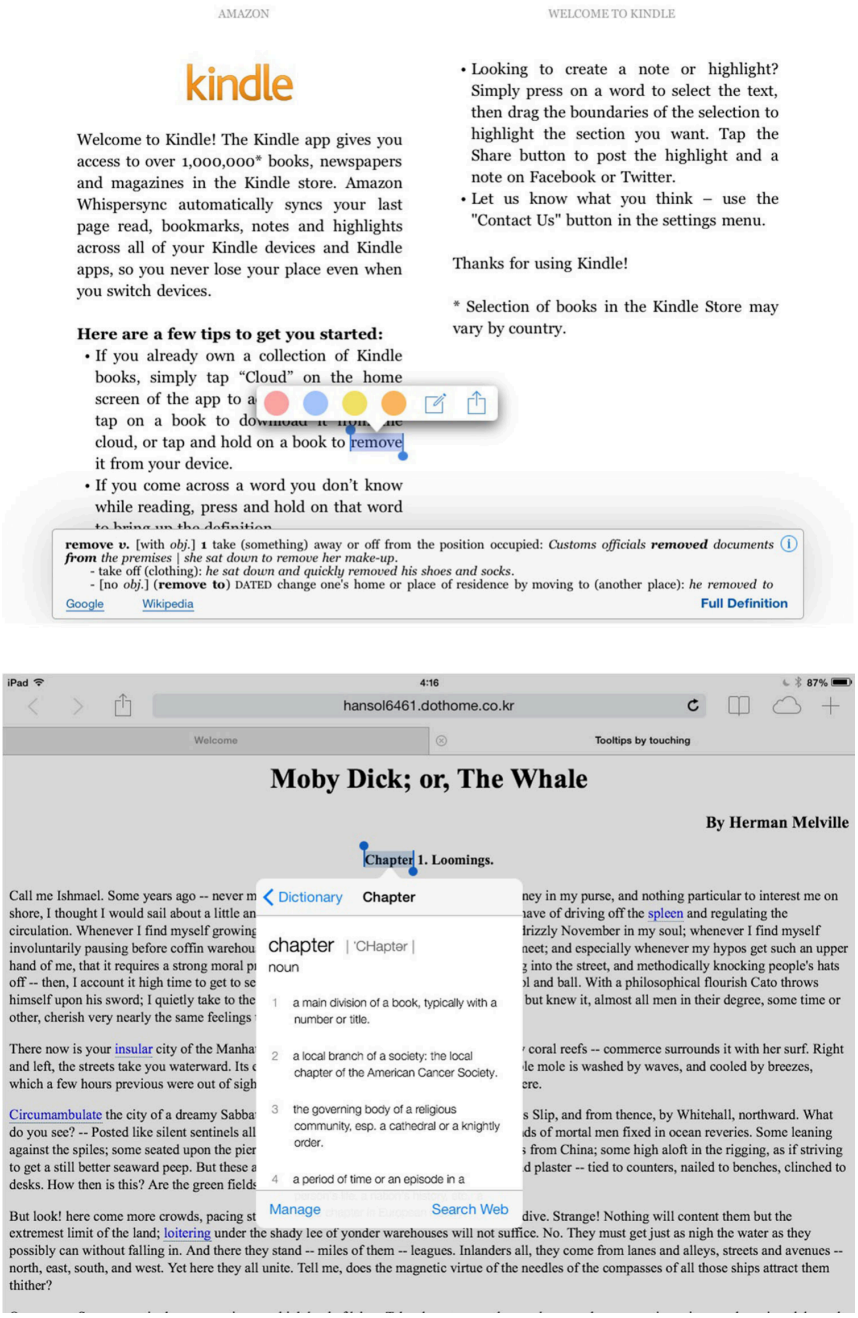


Fig. 1 Dictionary function pages on Kindle (*top*) and iPad Safari (*bottom*) screens

The Effects of Different Glossary Information

Inferring the meaning of an unfamiliar word from its context is considered a valuable learner strategy by researchers and students alike (e.g., Fraser 1999; Schmitt 1997). However, it has often been criticized due to the fact that learners often make incorrect inferences (Nassaji 2003). It is noteworthy that most discussions of the ineffectiveness of inferencing for FL vocabulary acquisition have been concerned with the usefulness of a *single* context surrounding a target word. Recently, the value of inferencing has been revisited, influenced by corpus linguistics and following the earlier work of researchers such as Johns (1986) and Cobb (1997), who advocated the use of concordance data to help learners effectively infer the

meaning of a target word, by offering them multiple instances of this word occurring in a wide range of sentential contexts. Some underlying assumptions here are that learners would be less likely to make incorrect inferences through the use of different contexts for target words, and retrieval of these words would be facilitated, as their encounters with the words would be naturally contextualized.

Nation (2009) embraces these ideas, and suggests that concordance data be used as a resource in computer-assisted reading. However, whether concordance lines are as effective as dictionary-type glossary information remains largely understudied. The only exception is Poole's study (2012), one of the first attempts to examine the value of concordance lines provided in glossing formats. In his study with English as a second language (ESL) students enrolled in a composition course at a U.S. university, he compared concordance glosses and dictionary glosses, which respectively referred to multiple example sentences including a glossed word and the dictionary definition of the glossed word. The results show that these two types of glossary information led to comparable gains in terms of the development of learners' vocabulary knowledge. However, it is questionable whether the sentences included in concordance glosses in this study can still be considered examples of 'concordance,' as they were modified on a syntactic level by the author. One is also left with the question of whether using *authentic* sentences as a method of inferring the meaning of unfamiliar vocabulary would be as effective as dictionary definitions. The present study aims to extend Poole's study by exposing learners to authentic concordance sentences, and comparing their effect with the two different formats of glossing mentioned in the previous section.

Cognitive Load Theory and Glossing

While glossing research in general is not theoretically grounded (but rather is driven by the need to provide pedagogical implications), some studies (e.g., Yeung 1999) have adopted a 'cognitive load theory' as their framework, which proposes that the process of learning can either be facilitated or hindered as a function of one's capacity for working memory in a given learning task (Sweller et al. 2011). The theory introduces the 'split-attention' effect, which "occurs when learners are required to split their attention between at least two sources of information that have been separated … spatially" (Sweller et al. 2011, p. 111). It follows then that glossing which is spatially separated from a glossed word may hypothetically result in the 'split-attention' effect, and consequently increase the learners' cognitive load.

AbuSeileek (2011), instead of drawing on cognitive load theory directly, interpreted his findings based on what he termed the 'principle of proximity'—reminiscent of the 'split-attention' effect, which proposes that the degree of proximity of a glossed word to glossary information has a bearing on the effectiveness of glossing in terms of vocabulary acquisition.

Based on his finding that the ‘after-glossed-word gloss’ condition fared better than other conditions, he concluded that “[physically] the nearer the meaning of the gloss to the glossed word is … the higher achievement is obtained” (p. 1288), suggesting the principle of proximity. On the other hand, Chen and Yen (2013) only alluded to some notions surrounding cognitive load theory, and did not discuss their findings in this framework.

The present study adopts the cognitive load theory more directly in order to examine whether the three glossing types mentioned would result in similar or different amounts of cognitive load on the part of the participants, using measurements of cognitive load that have been widely administered in previous studies.

Research Questions

In light of the review of the literature above, the present study proposes two research questions as follows:

(1) Do EFL adult learners assigned to different glossing types in hypermedia reading environments show different gains in terms of the short-term and long-term word recall?

Hypothesis 1a: The gains of the group assigned to the concordance glossing condition will be lower than those of the groups assigned to the definition-type of glossing conditions.

Hypothesis 1b: The gains of the group assigned to the tooltip-type glossing will be higher than those of the group assigned to the frame-type glossing.

(2) Do EFL adult learners assigned to different glossing types in hypermedia reading environments show differences in their self-perceived amount of cognitive load and in their perception of the usefulness of the respective glossing types they encounter?

Hypothesis 2a: In view of their level of English language proficiency, the group assigned to the concordance glossing condition and the other two groups assigned to definition-type glossing will experience a similar amount of cognitive load.

Hypothesis 2b: The groups assigned to definition-type glossing will rate the usefulness of their glossing type more positively, as definition-type glossing is more familiar to them.

Research Design

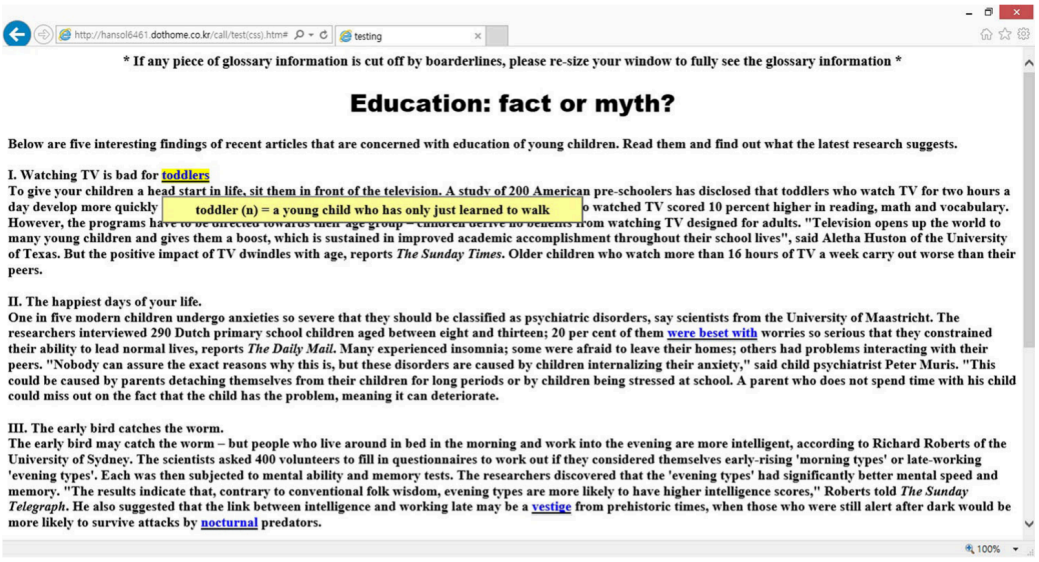
Participants

The group of participants in the present study was made up of 83 Korean first-year university students, from a wide range of majors. All of them had about 9 years' experience learning English in formal school contexts, and reported themselves to be competent users of computers. The average score of the participants on the test of English for international communication (TOEIC) was approximately 720. They were enrolled in a mandatory course entitled "English Reading." The participants were distributed among six classes, all of which shared the same textbook and curriculum, but which were taught by three different instructors.

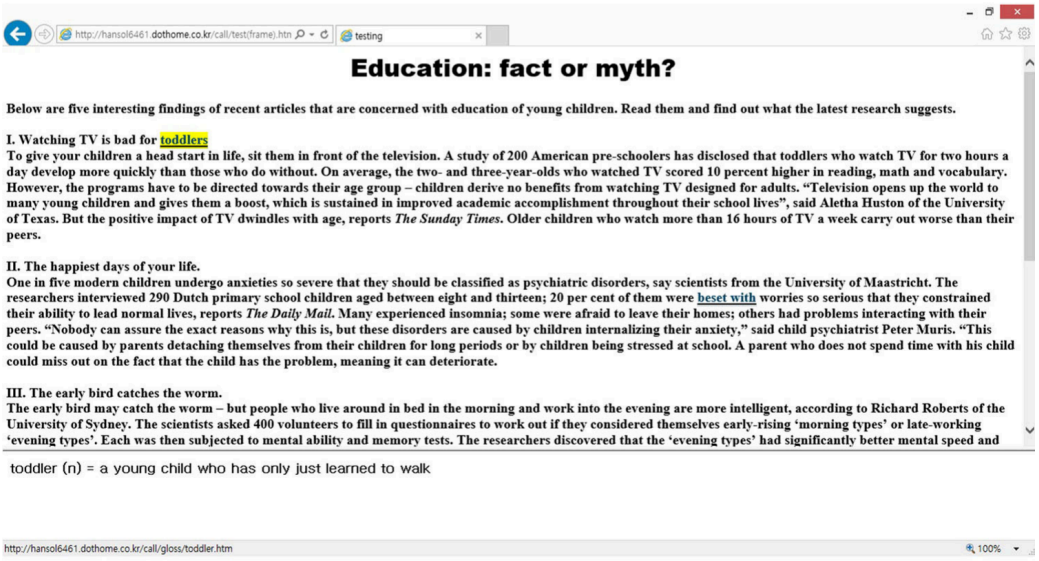
Material and Experimental Conditions

The present study used a text entitled "Education: Fact or Myth?" (893 words) extracted from *Cutting Edge Advanced with Phrase Builder* (Cunningham et al. 2003). This text was selected because it was found to be suitable for the aforementioned course in terms of its syntactic level and length, as judged by the instructors of this course. The text also included some difficult words that would make it ideal for measuring participants' learning of target vocabulary. The authors additionally replaced some of the words in the text with less frequent English words, and consequently chose 30 potential target English words from the text based on a pilot study involving eight students similar in profile to the participants in the main study. Then, based on the results of the pre-test of vocabulary, they selected ten words as the final target words to be glossed (see *Procedure* for more details).

The authors then developed three different versions of hypermedia reading texts using tooltip-type glossing (TG), frame-type glossing (FG), and concordance glossing (CG). The first two versions provided English definitions of target words in the two different formats of electronic textual glosses (see Fig. 2 for their user interface designs). In the tooltip-type condition, a pop-up box showing the definition of a glossed word appeared temporarily when the participant moved the mouse cursor over the glossed word, and it disappeared when he or she moved the cursor away from the word. This glossing format was designed in such a way that it would not obscure any contexts surrounding the glossed word. On the other hand, in the frame-type condition, the definition appeared in the bottom frame of the screen when the participant clicked the glossed word, and it remained visible until he or she clicked another glossed word.



Tooltip-type glossing (TG)



Frame-type glossing (FG)

Fig. 2 Screenshots of the two formats of glossing

In the CG version, the glossing device was equipped with concordance sentences involving the glossed words, through which participants were given authentic sentences from two authentic reference corpora (the "British National Corpus" and "Brown"). The concordance sentences for the glossed words were carefully selected so that the learners would be able to infer the correct meaning of target words in accordance with the context, and they were later piloted with the students mentioned above.

The interface design of this condition was the same as that of the FG format. Concordance sentences were given in the frame-type rather than tooltip-type format, as the pilot study showed that the latter blocked too large a portion of surrounding contexts due to the size of the

text. For each glossed word, three authentic sentences were provided, in consideration of the results of the pilot study that indicated three concordance sentences were most appropriate both time-wise and cognitively for the learners at this proficiency level (see Fig. 3).

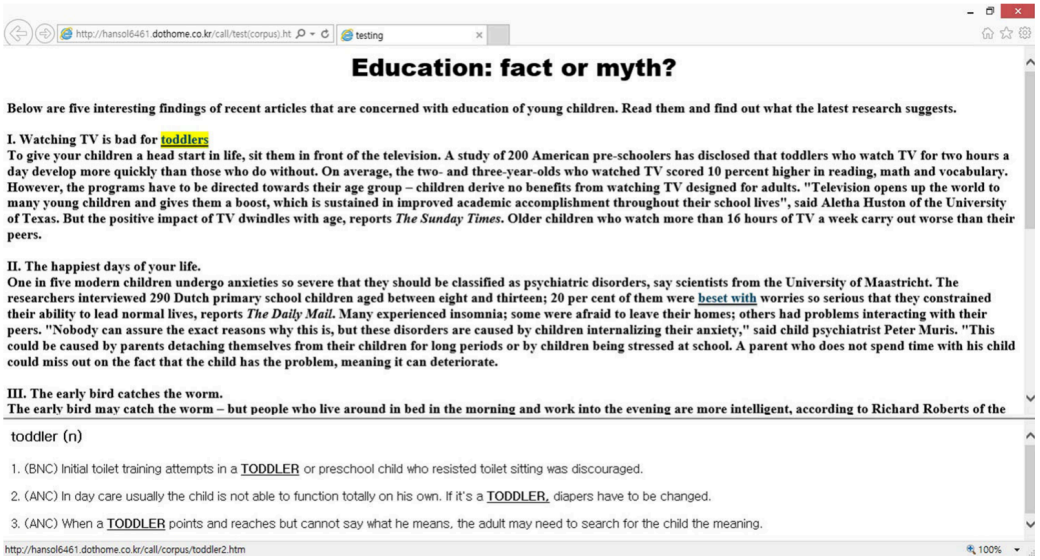


Fig. 3 Screenshot of concordance glossing (CG)

Testing and Post-reading Questionnaire

In the present study, a passive recall test of vocabulary was used (see Appendix 2). With this test, the participants were asked to "demonstrate the understanding of the meaning of the L2 [second language] word" (Laufer and Goldstein 2004, p. 406), by asking them to provide the meaning of the ten target words (either in their L1 or English).

Adapting the questionnaire in Yeung (1999), the authors developed an online post-reading questionnaire which asked about the participants' perception of the difficulty of the reading text and target vocabulary, which was in turn operationalized as the mental effort (i.e., cognitive load) involved in dealing with them via the respective glossing devices. These two items were constructed on a 7-point Likert scale, with one being very difficult and seven being very easy. It also included the perceived usefulness of the glossing devices the participants received, measured on a 7-point Likert scale with one being completely useless and seven being extremely useful. Lastly, the questionnaire included open-ended questions asking their opinions of the respective glossing devices they were provided with.

Procedure

At the beginning of the study, the participants were told that their performance during the experiment would not in any way affect their grades in the course "English Reading." The six classes in which the participants were taught were randomly distributed on a class level to one

of the three glossing conditions that the authors had set up.

The participants were all given the pre-test of vocabulary, which consisted of 30 words found in the target text. The purpose of the pre-test was first to finalize target words that were unfamiliar to the participants, and second to examine whether there were any significant differences among the three groups allocated to the different conditions in terms of their knowledge of the target words.

The main reading activity was implemented 2 weeks after the pre-test. Each pair of the two classes allocated to the same learning condition completed the reading activity in the same computer lab, without the presence of the classes allocated to the other conditions. Instruction was given prior to the reading activity as to how to activate glossing devices in their respective hypermedia conditions. The participants were asked to read the text for 25 min, and were not notified of the upcoming vocabulary test. They were encouraged to use the glossing devices in order to comprehend the text and target words. While the participants engaged in reading, the instructors ensured that the participants used the glossing devices associated with the glossed words in comprehending unfamiliar target words. Also by using visual log files, it was confirmed that every participant looked up the glossary information or concordance sentences of all the glossed words.

After 25 min, the participants were asked to close the web page on which the reading text was displayed. Subsequently, they were given the post-test of vocabulary and were asked to recall the meaning of the target words within 3 min. This test was equivalent to the pre-test of vocabulary in its format, but only included the finalized set of the target words. After the post-test, the instructors asked the participants to complete the online post-reading questionnaire. Two weeks later, the participants were given the delayed test of vocabulary without any prior notice.

Scoring

For the vocabulary recall test, two points were given for the completely correct meaning of each target word, and one point for the partial meaning. For example, the complete answer for the target word ‘plethora’ was “a very large amount, an amount that is much greater than what is necessary” as given in the glossing for the TG and FG conditions. In cases where the participant provided an answer such as ‘many’ which lacks the nuance of ‘excessiveness,’ one point was given, as it was a partially correct answer. One of the authors and another instructor teaching the same course graded the vocabulary tests individually, and 20 % of the total tests were scored together for an inter-rater reliability check. Cohen’s Kappa was 0.94 ($p = 0.01$), which indicated a high degree of consensus between the raters.

Data Analysis

For the first research question, a mixed ANOVA with two main factors, *Time* (i.e., pre-test, post-test, and delayed test) and *Condition* (i.e., tooltip-type, frame-type, and concordance), was conducted. Furthermore, subsequent analyses of each factor were conducted: (1) paired-samples *t* tests were conducted to further examine the effect of *Time* for each group and (2) a one-way ANOVA was conducted for the overall comparison of the three groups in terms of their scores on the post-test, and delayed test of vocabulary, along with post-hoc Scheffé tests. A Bonferroni correction (0.05/9) was applied to multiple paired-samples *t* tests (three tests for each group). For the second research question which concerned the questionnaire items, the participants' responses were compared descriptively. Although there was an overall total of 83 participants in the study, only 79 questionnaires were analyzed, as four participants failed to submit their responses to the online questionnaire due to technical issues.

Preliminary Analyses of the English Proficiency Test and Pre-test of Vocabulary

The results of a one-way ANOVA on the participants' TOEIC scores and the pre-test of vocabulary did not reach a statistical significance (F (2, 80) = 0.12, p = 0.89 for TOEIC and F (2, 80) = 0.23, p = 0.79 for the pre-test), indicating that there was no significant difference among the three groups in their general English proficiency and their knowledge of the target words prior to reading at the outset of the study. It was found in the analysis of the pre-test that the participants had on average partial knowledge of only one target word before the experiment.

Results

Results of Vocabulary Recall Tests

Table 1 below provides descriptive statistics for the scores of the three groups on the vocabulary recall test at three testing periods.

Table 1
Means and standard deviations for the recall tests and the results of *t* tests comparing each group's performance at three testing periods

Group	**Pre-test (max = 20.0)**	**Post-test (max = 20.0)**	**Delayed test (max = 20.0)**	***t* tests**	
	Mean (*SD*)	**Mean (*SD*)**	**Mean (*SD*)**		***t* value**
TG (n = 29)	0.83 (0.85)	7.24 (3.24)	7.03 (4.06)	Pre < Post	9.98*
				Pre < Delayed	7.84*
				Delayed ≒ Post	0.27
FG (n = 26)	0.92 (0.98)	8.92 (4.14)	7.96 (4.11)	Pre < Post	9.45*
				Pre < Delayed	8.39*
				Delayed ≒ Post	2.35

Group	Pre-test (max = 20.0)	Post-test (max = 20.0)	Delayed test (max = 20.0)	*t* tests	
	Mean (*SD*)	Mean (*SD*)	Mean (*SD*)		*t* value
CG (*n* = 28)	0.75 (0.97)	4.57 (2.74)	4.00 (2.74)	Pre < Post	6.67*
				Pre < Delayed	6.06*
				Delayed ≒ Post	1.65

* $p < 0.006$ (*Bonferroni* correction applied)

As stated in the data analysis section, the participants' scores on the recall test were subjected to a mixed ANOVA with two main factors, *Time* and *Condition*, and the results are summarized in Table 2.

Table 2
Results from a mixed ANOVA with *Time* and *Condition* factors

Factors	Correction	df	*F*	Sig.	Partial η^2	Observed power
Time (Pre-test, post-test, and delayed test)	Greenhouse-Geisser	1.83, 146.40	151.69	0.001	0.66	1.00
Condition (TG, FG, and CG)		2, 80	12.27	0.001	0.24	0.99
Interaction between *time* and *condition*	Greenhouse-Geisser	3.66, 146.40	42.56	0.001	0.14	0.98

This analysis revealed that there was a statistical main effect for *Time* using the Greenhouse-Geisser correction, F (1.83, 146.40) = 151.69, $p = 0.001$ with a large effect (partial $\eta^2 = 0.66$). It was also found that the *Condition* effect was statistically significant, F (2, 80) = 12.27, $p = 0.001$, again with a large effect size (partial $\eta^2 = 0.24$). Lastly, the analysis showed that there was a statistical effect for interaction between *Time* and *Condition* using the Greenhouse-Geisser correction, F (3.66, 146.40) = 42.56, $p = 0.001$. This means that the participants from the three groups did not perform in a parallel manner across the three testing points, and this statistical significance appears to be caused by the fact that the mean score of the CG group on the post-test did not increase as sharply as those of the FG and TG groups. This interaction was also observed to show a large effect size, and a sufficiently high power to find statistical differences.

To give more details on the main effect of *Time*, paired-samples *t* tests for each group with Bonferroni correction revealed significant differences in their mean scores between the post-test and pre-test as well as between the delayed test and pre-test (see Table 1 for details). For all three groups, there was no significant difference in their mean scores between the post-test and delayed test, indicating no significant attrition of this knowledge.

A one-way ANOVA was also performed to determine which pair of conditions resulted in significant differences, if any, for the post-test and delayed test. In the case of the post-test, there was a significant group difference, F (2, 80) = 11.27, $p = 0.001$. Comparisons using a post-hoc test confirmed that the mean scores of the FG and TG groups were significantly higher than that of the CG group, but the difference in the mean scores of the FG and TG

groups did not reach statistical significance. A similar result was found with the delayed test, with the results of ANOVA again revealing the statistical significance of the differences among the groups (F (2, 80) = 8.64, p = 0.001). Post-hoc analysis indicated that the scores of the FG and TG groups were higher than that of the CG group on this test, but again no significant difference between the scores of the FG and TG groups was found. Figure 4 presents the developmental patterns of vocabulary recall knowledge for the three groups.

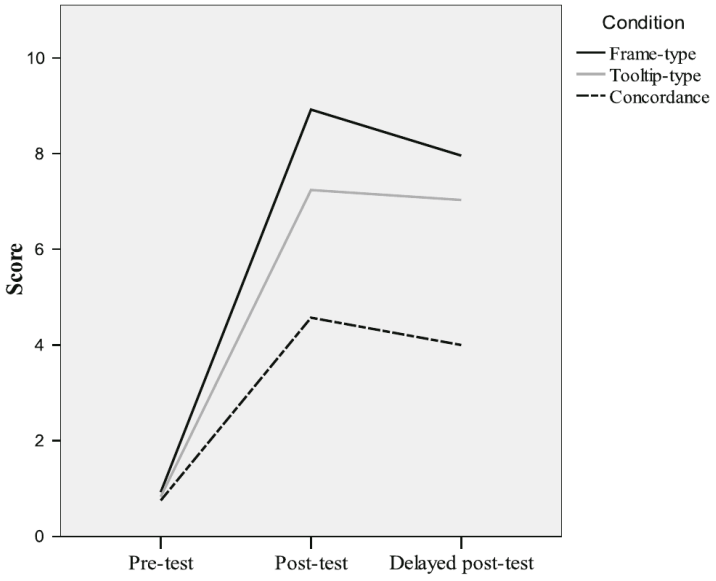


Fig. 4 Developmental patterns of vocabulary knowledge for the three groups

Results of Questionnaire Items

Descriptive statistics concerning the participants' effort scores for reading and comprehension of target vocabulary via different glossing devices are given in Table 3.

Table 3
Means and standard deviations of the three groups' responses to the difficulty of the reading text and target vocabulary

Group	Effort on comprehension[a] Mean (*SD*)	Effort on vocabulary[a] Mean (*SD*)
TG (n = 27)	3.26 (1.10)	2.93 (0.92)
FG (n = 25)	3.08 (0.95)	2.84 (0.94)
CG (n = 27)	3.00 (1.18)	2.44 (1.01)

[a] *1* very difficult, *2* difficult, *3* slightly difficult, *4* moderate, *5* slightly easy, *6* easy, *7* very easy

The results show that the groups were more or less equivalent in terms of their perception of the difficulty of the reading text and target words. On average, the participants found the target text to be slightly difficult, while they perceived the level of the words found in this text to be

more cognitively challenging than the level of the text itself. The mean score of the perception of the difficulty of vocabulary for the CG group was lower than those of the other two groups, but only to a small degree.

Another questionnaire item dealt with the participants' perception of the usefulness of the electronic glossing devices that they were provided with in comprehending unfamiliar FL words, again in the form of the Likert scale. Among the three groups, the concordance group rated their glosses lowest (M = 3.56, SD = 1.58), whereas the groups who were given the definitions of glossed words rated their glosses more positively than the concordance group (M = 5.07, SD = 1.27 for the tooltip-type group and M = 4.88, SD = 0.97 for the frame-type group).

Discussion and Conclusion

The present study investigated whether three different types of electronic glossing in hypermedia environments could facilitate the development of vocabulary knowledge for university-level EFL learners, and whether there would be any differences among three glossing types in terms of differences in the participants' recall knowledge of target vocabulary. The findings revealed that the three groups all made significant gains after reading the target text with their respective glossing types, and further revealed that these gains were retained for 2 weeks. The development of vocabulary recall knowledge for these three groups should be interpreted with some caution, as the present study did not include a pure control group. However, in light of the general findings from previous glossing research that glossing almost without exception results in better learning than the lack thereof, it can be suggested that the three glossing types in the present study enhanced the participants' vocabulary knowledge.

Regarding the differences among the three glossing types in the post-test and delayed tests, the results showed that those who read the text with definitions of target vocabulary in tooltip-type and frame-type glosses fared better than those who had to infer the meaning of glossed words from authentic concordance sentences; thus, the authors' hypothesis regarding the relative effectiveness of concordance glossing and definition-based glossing was supported. The findings here are in line with those of previous FL corpus studies (e.g., Fraser 1999), where inferring the meanings of target words without subsequent confirmation of these meanings resulted in a lower retention rate than inferencing followed by confirmation. The findings of the present study did not match those of Poole (2012), where meaning-inferred and meaning-given information were found to produce similar gains in vocabulary knowledge. Perhaps this discrepancy may be attributed to the nature of the sentences provided in Poole's study. The sentences given to the participants in his study were syntactically modified by the researcher,

and perhaps were more effective than those in the present study in terms of facilitating participants' inferencing of vocabulary meaning. It was found from the participants' answers to the open-ended questions that the concordance group generally believed that the concordance sentences they were exposed to were rather difficult to process, and thus to use this information in inferring the meaning of target words was not a successful strategy for them.

In reference to the findings related to the post-reading questionnaire, the mental effort spent on target vocabulary items was found to be nearly equivalent among the three groups, and thus the authors' hypothesis that the three groups would experience a similar amount of cognitive load was confirmed. This finding, on the other hand, implies that tooltip-type and frame-type glosses resulted in greater gains in this type of vocabulary knowledge than concordance glosses, while maintaining a more or less equivalent amount of cognitive load on the learners. Thus, the available evidence appears to suggest that tooltip-type and frame-type glosses were more efficient than concordance glosses. Perhaps another explanation for the difference in the gains in vocabulary recall between the concordance and the other two groups comes from a questionnaire item concerning the participants' perception of the usefulness of the respective glossing types they received. To this questionnaire item, the two groups exposed to the definition of target vocabulary provided more positive responses than the concordance group. Considering the fact that this questionnaire was administered immediately after the reading activity, these rather negative responses from the concordance group may have reflected their disappointment in this glossing type in terms of comprehending the meanings of target words.

On the other hand, the finding of no significant difference between tooltip-type and frame-type glossing is in accordance with Chen and Yen (2013), where different glossing formats resulted in rather similar gains in terms of the learning of target vocabulary. However, the finding in favor of after-glossed-word glossing and marginal glossing over bottom glossing in AbuSeileek (2011) was not echoed in the present study, with the authors' hypothesis not being supported; consequently, the 'principle of proximity' was not corroborated by the data here. One postulation is that the participants in the present study were self-reported competent users of computers, and were not therefore disadvantaged by an 'attention split' between the glossed words and their relevant glossary information. Whether there is any threshold level at which this split-attention effect fades away requires future studies, and these should aim to sample participants with different competencies in the use of computers.

In light of the cognitive load theory, the existence of no significant difference between the frame-type and tooltip-type groups is supported by another finding that concerns the mental efforts induced by each glossing device and the participants' perceptions of the usefulness of

each glossing format: the frame-type and tooltip-type groups were found to have experienced a similar amount of cognitive load as a result of their respective electronic glossing types, which they also rated similarly in terms of their usefulness.

There are some limitations of the present study, two of which will be discussed below. First, adopting more diverse types of vocabulary measurements may have captured positive effects of concordance glosses on the development of other aspects of vocabulary knowledge. While such an undertaking would have resulted in informative findings, there were some time constraints placed on the authors to administer the reading task, vocabulary measurement, and post-reading questionnaire over a given class period, and consequently they had to focus on measuring the participants' ability to retrieve and recall the meaning of target vocabulary—which was deemed one of the most important aspects to gain in their contexts. Second, as one reviewer rightly suggested, each participant was exposed to only one glossing type, and thus an overall finding on their perception of each glossing type may need to be read with some caution.

Based on the findings here, research in the future should address two major issues as follows. First, future studies should include a more diverse range in terms of learner population. While the participants in the study were around the intermediate level of English proficiency and had a considerable amount of experience with computer technology, whether the findings here can be generalized beyond the present population remains unknown. Second, they should also address the question of how to present different types of glossing information *in combination* in terms of the order of this presentation or the physical glossing format for each type of glossary information. The findings of the present study, along with previous studies that have examined the effects of corpora use (but not in a glossing format) have hinted that providing concordance lines only may not be the most effective route in terms of learning the meaning of target vocabulary. However, this does not necessarily mean that it has little pedagogical value. Concordance sentences, if presented along with the definition-type of glossary information as an aid to confirm learners' inferences, may demonstrate more positive learning outcomes.

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Appendix 1: Details on Tooltip-type and Frame-type Glosses

Details on tooltip-type and frame-type glosses in light of previous research are as follows: Tooltip-type glosses share a thread with the integrated format in Yeung (1999) in that they always appear physically close to the glossed words. This makes tooltip-type glosses also similar to after-glossed-word glosses in that they take advantage of their close proximity to the

glossed words (AbuSeileek 2011). However, tooltip-type glosses have an additional advantage over after-glossed-word glosses. The latter insert glossary information directly into the target sentence containing a glossed word when a user clicks the word, and this process appears to cause some changes in the shape of the surrounding paragraph and sentences. This may then bring about negative effects on one's reading flow. Contrary to this format, tooltip-type glosses were designed not to affect the structure of a target text (see *Material and Experimental Conditions* for more details in terms of its design). Frame-type glosses, on the other hand, are largely equivalent to bottom glosses in AbuSeileek (2011), as the glossary information appears in the bottom frame of the screen, once it is activated. This type of glossing may be preferred by learners who do not want glossary information to appear immediately around the glossed word.

Appendix 2: Meaning-recall Test

Provide the meaning of the following words in either English (synonym/definition) or Korean.

Plethora	_____
Attest	_____
Be beset with	_____
Corrode	_____
Dismayed	_____
Endowments	_____
Idiot savant	_____
Nocturnal	_____
Toddler	_____
Vestige	_____

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